Lesson 21 One-to-one and Onto Functions

Questions 1 to 12:

a) determine if the given function is one-to-one, onto, or both, with proof
b) if it is not onto, give its image
c) if it is a bijection, give its inverse (do this part after Lesson 22)

1. $f: \mathbb{Z} \to \mathbb{N} \cup \{0\},\ f(x) = |x|$
2. $f: \mathbb{N} \to \mathbb{R},\ f(x) = \ln x$
3. $f: \mathbb{R} \to (0, \infty),\ f(x) = e^x$
4. $f: \mathbb{R} \to \mathbb{R},\ f(x) = x^2 + x$
5. $f: \mathbb{R} \to [-1,1],\ f(x) = \sin x$
6. $f: [0, \infty) \to \mathbb{R},\ f(x) = e^{-x^2}$
7. $f: \mathbb{R} \to \mathbb{Z},\ f(x) = \lfloor x \rfloor$
8. $f: \mathbb{R}\backslash\{0\} \to \mathbb{R},\ f(x) = 1/x$
9. $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z},\ f(n, m) = n - m$
10. $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z},\ f(x, y) = xy$
11. $f: \mathbb{R} \times \mathbb{R} \to \mathbb{Z},\ f(a, x) = \begin{cases} 0 & \text{if } x < a \\ 1 & \text{if } x \geq a \end{cases}$
12. $f: [0, \infty) \to (0,1]$ defined by $f(x) = 1/(1 + x^2)$

13. Let $f: \mathbb{Z} \to \mathbb{Z}$ be a one-to-one function. Prove the following:
    a) Let $g: \mathbb{Z} \to \mathbb{Z}$, $g(x) = 2f(x)$. Then g is one-to-one.
    b) Let $g: \mathbb{Z} \to \mathbb{Z}$, $g(x) = f(x + 1)$. Then g is one-to-one.
14. Let $f: \mathbb{R} \to \mathbb{R}$ be an onto function. Prove:
    a) Let $g: \mathbb{R} \to \mathbb{R}$, $g(x) = 2f(x)$. Then g is onto.
    b) Let $g: \mathbb{R} \to \mathbb{R}$, $g(x) = f(x) - 1$. Then g is onto.